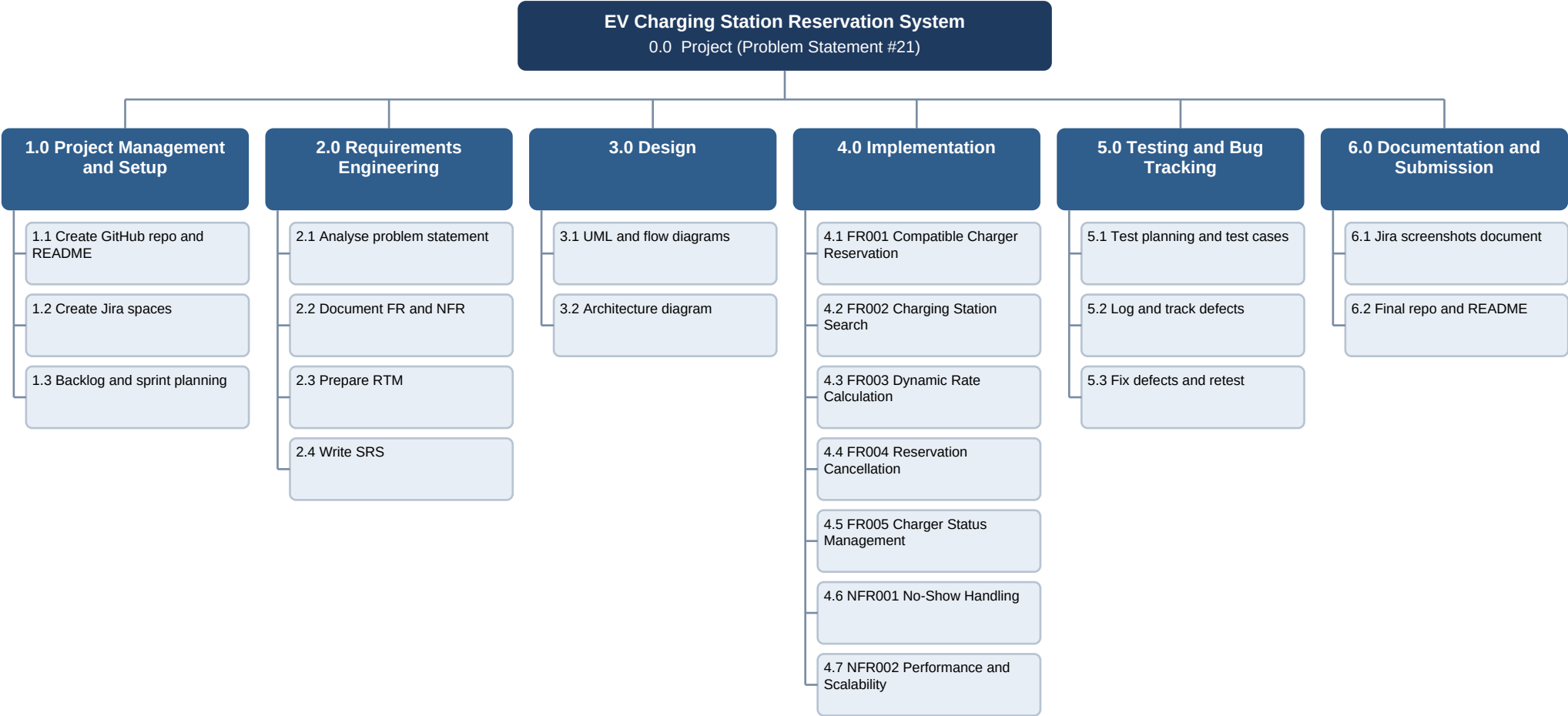


Work Breakdown Structure (WBS)

EV Charging Station Reservation System | Problem Statement #21 | Shriya S R, PES1UG24CS449, 5 H | Version 1.0, 02 October 2026



WBS Dictionary

Each work package is listed with its deliverable, the Jira items that track it, and an effort estimate. For implementation work packages effort is **0.5 person-day per story point (assumed)**. All other estimates are the author's planning estimates.

WBS ID	Work package	Deliverable	Jira reference	Effort (person-days)
1.0	Project Management and Setup			2.5
1.1	Create GitHub repository and README	Repository with folder structure and README	GitHub	0.5
1.2	Create Jira spaces (Scrum, Kanban, Bug Tracking)	Three Jira spaces for Problem Statement #21	Jira	1
1.3	Backlog creation and sprint planning	Epics, 7 stories, tasks, subtasks and a sprint (30 Aug to 11 Sep 2026)	SBP-1 to SBP-30	1
2.0	Requirements Engineering			4
2.1	Analyse Problem Statement #21	List of requirements and actors	-	1
2.2	Document functional and non-functional requirements	FR and NFR document	FR001-FR005, NFR001-002	1
2.3	Prepare Requirements Traceability Matrix	RTM table	-	0.5
2.4	Write Software Requirements Specification	SRS document (PDF)	-	1.5
3.0	Design			2.5
3.1	Use case, alternate flow and exception flow diagrams	UML, alternate flow and exception flow diagrams	-	1.5
3.2	Architecture diagram	Architecture diagram with explanation	-	1
4.0	Implementation			23
4.1	FR001 Compatible Charger Reservation (8 SP)	Connector matching, slot creation, connector type validation	SBP-1, 17, 18, 31	4
4.2	FR002 Charging Station Search (5 SP)	Station search and availability filtering	SBP-2, 19, 20	2.5
4.3	FR003 Dynamic Power Rate Calculation (5 SP)	Rate calculation and rate display	SBP-3, 21, 22	2.5
4.4	FR004 Reservation Cancellation (3 SP)	Cancellation and release of slot	SBP-4, 23, 24, 38	1.5
4.5	FR005 Charger Status Management (5 SP)	Status updates and availability synchronisation	SBP-5, 25, 26	2.5
4.6	NFR001 No-Show Handling (8 SP)	No-show detection and automatic socket release	SBP-6, 27, 28	4
4.7	NFR002 Performance and Scalability (12 SP)	Peak-load test and scalability validation	SBP-16, 29, 30	6

WBS ID	Work package	Deliverable	Jira reference	Effort (person-days)
5.0	Testing and Bug Tracking			4.5
5.1	Test planning and test case design	Test cases traced to requirements	-	1.5
5.2	Log and track defects	Seven defects in the Bug Tracking space	BP-1 to BP-7	1
5.3	Fix defects and retest	Fixed code and regression test results	BP-1 to BP-7	2
6.0	Documentation and Submission			1
6.1	Compile Jira screenshots document	Project Creation Screenshots (Word file)	-	0.5
6.2	Organise repository folders and final README	Final submission repository link	-	0.5
	Total effort			37.5

Effort Summary by Phase

Phase	Effort (person-days)	Share of total
1.0 Project Management and Setup	2.5	7%
2.0 Requirements Engineering	4	11%
3.0 Design	2.5	7%
4.0 Implementation	23	61%
5.0 Testing and Bug Tracking	4.5	12%
6.0 Documentation and Submission	1	3%
Total	37.5	100%

Assumptions and Dependencies

- Phase 2 (Requirements) must be finished before Phase 3 (Design) and Phase 4 (Implementation) start.
- Phase 5.1 (test design) can start once the SRS acceptance criteria are written, in parallel with Phase 4.
- Story points (total 46: FR001 8, FR002 5, FR003 5, FR004 3, FR005 5, NFR001 8, NFR002 12) come from the Jira Scrum backlog.
- Sprint 1 in Jira runs from 30 August 2026 to 11 September 2026.

- The project is carried out by one student, so all effort is on a single resource.